Eric Feng

Principal/Lead Engineer | Unreal Engine Specialist |
Ex-Activision, ex-EA, ex-Wētā FX, ex-ILM

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With over 20 years of dedicated experience in the software industry, I am a passionate programmer whose work has been widely embraced by top-tier game studios, leading visual effects houses, and renowned digital content creation platforms.

WORK EXPERIENCE

Lead Software Engineer
The LEGO Group

12/2024 - Present — *Billund, Denmark*

Achievements/Tasks

- Lead development of the LEGO Digital Brick SDK on Unreal Engine, enabling authentic LEGO brick-building experiences within the UEFN ecosystem.
- Spearhead a cross-functional team to deliver intuitive digital building UX, high-fidelity brick physics, character animation systems, and performance-optimized tooling for creators.

Principle Gameplay Engineer
Self-Employed

06/2022 - Present — *Remote*

Achievements/Tasks

- Independent Game Development Consultant specializing in gameplay systems and animation.
- Collaborate closely with design and animation teams to architect and implement advanced gameplay mechanics, with a focus on character traversal, 3C (Character, Camera, Controls), and AI behavior.
- Deliver high-impact solutions that elevate player experience and technical fidelity across diverse projects.

Expert Gameplay Engineer
Sledgehammer Games, Activision

11/2021 - 05/2022 — *Melbourne, Australia*

Achievements/Tasks

- Research and implement gameplay mechanics for up-coming Call Of Duty games.
- Optimize loot pickup and destruction system with ECS framework.

Software Engineer
Frostbite - EA

03/2020 - 10/2021 — *Melbourne, Australia*

Achievements/Tasks

- Main developer at the core team of Entity-Component-System (ECS) framework in Frostbite Engine.
- Collaborate in features development with different game teams around the globe.

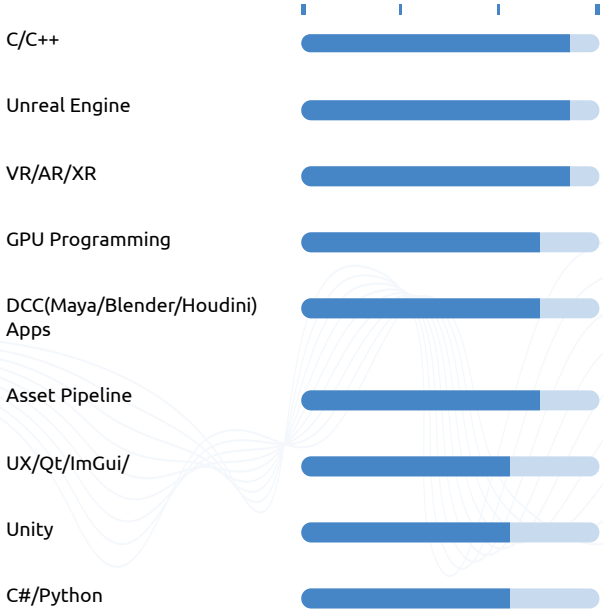
Senior Software Developer
Wētā FX

08/2013 - 03/2020 — *Wellington, New Zealand*

Achievements

- Designed and developed multiple award-winning real-time interactive VR/ AR/ XR applications in Unreal Engine and Unity with a focus on gameplay, character animation, rendering and performance profiling.
- Main developer for an academy award-winning scene description system (Atlas) focusing on low-level API design, animation, evaluation system and real-time rendering for the proprietary scene data.

TECHNICAL SKILLS



PROJECTS

Lego Brick Editor in Unreal Engine / UEFN (12/2024 - 04/2026)

- A collaboration project with Epic Games. It is used in LEGO Fortnite and UEFN.
- Lead Software Engineer. Have been working on user experience of digital brick building, animations, physics and optimization.
- Support both internal and external vendors for using Atom framework in UE5 projects.

Exodus (04/2023 - 12/2024)

- Mostly focused on player traversal gameplay mechanics and gameplay animation implementations.
- Product owner and main contributor for swing, grapple, balance beam and camera transition systems.
- Ensured the gameplay systems work well with GAS, companion systems and NPC interactions.

Call of Duty: Modern Warfare III (12/2021 - 05/2022)

- Created a new Point-Of-Interest (POI) for NPCs.
- Adopted ECS system for gameplay system optimization.
- Underwater gameplay animation setup.

Towerborne (11/2020 - 12/2021)

- Co-development consultant on Stoic's action-RPG, built in Unreal Engine.
- Ported the game to Xbox from Windows.
- Set up the Epic Online Services subsystem integrated with PlayFab.
- Performance tuning for character spawning, garbage collection, and game load times to hit the framerate target on all supported platforms

Frostbite Engine (03/2020 - 10/2021)

- Design and develop foundation modules for Entity-Component-System (ECS) in Frostbite engine.
- Collaborate with tech artists to define new workflows for FX, Lighting and Animation.

R&D Engineer
Industrial Light & Magic

09/2011 - 07/2013 *Singapore*

Achievements

- Designed and developed academy Sci-tech award-winning ILM's proprietary procedural rigging system, Block Party 2.
- Developed HoudiniBatch, a render farm manager for Houdini allowing FX artists to maximize the efficiency of the academy award-winning ObaQ render queue management system.

Motion Capture Pipeline Technical Director
Framestore

09/2010 - 09/2011 *London, UK*

Achievements

- Built up the whole motion capture data pipeline and production tools from scratch, including data repository management, in-house Motionbuilder real-time control panel design/develop and Vicon mocap processing pipeline.
- Worked on stage to provide technical supports for multiple shows including Gravity (2013), Wrath of Titan (2012) and etc.

Software Engineer
Autodesk

12/2007 - 08/2010 *Shanghai, China*

Achievements

- Designed, developed and maintained Autodesk FBX Library (includes SDK and plug-ins) , Autodesk Material Library and Autodesk Mudbox
- Collaborated with multiple Autodesk product teams in different countries on FBX pipeline
- Assisted high profile clients in resolving technical issues.

Childish Gambino's Pharos (09/2018 - 12/2018)

- 2019 VES Award for Outstanding Visual Effects in a Special Venue Project.
- Main developer for Dome projection VR preview, gameplay and interactive graphics for the concert.

Atlas (08/2013 - 08/2016)

- Wētā FX's proprietary scene description and evaluation framework; main developer of the project.
- 2024 Academy® Scientific and Technical Award Technical Achievement.
- Designed and maintained a binary-compatible API.
- Designed, implemented and optimized the evaluation framework.
- Ensured the C++ and Python APIs were well tested and documented.

Lucasfilm Block Party 2 (09/2012 - 06/2013)

- Lucasfilm's in-house, Maya-based procedural rigging system, successor to the original Block Party (Smith and White, 2006). Lets expert and novice riggers visually build shareable rigs from expertly designed rig pieces.
- 2018 Academy® Scientific and Technical Award Technical Achievement.
- Main designer and developer, covering the system back end and the front-end UX design and implementation.
- Built with Python, the Maya Python API, Qt and PyQt.

Framestore Motion Capture Production Tools (09/2010 - 09/2011)

- Built the entire motion capture data pipeline and production tools from scratch: data repository management, an in-house MotionBuilder real-time control panel and the Vicon mocap processing pipeline.
- All tools and plug-ins are cross-platform, written in Python, MEL, C++ and Qt.
- Provided on-set technical support for shows including Gravity and Wrath of the Titans.

Autodesk FBX (01/2008 - 08/2010)

- Main designer and developer of instancing, tangent-space export, subdivision surfaces, hardware shader (CgFX and HLSL) support, Autodesk Material Library data interpolation, the Unreal Engine 3 FBX importer/exporter and Single Step Suites interoperability.
- Designed and implemented FBX SDK features and plug-ins, mostly around polygon, material and scene graph technologies.
- Main contributor to FBX SDK and plug-in releases From 2009 to 2012.

EDUCATION

Computer Science (09/2002 - 07/2006)

Shanghai Jiao Tong University
Shanghai, China. GPA 3.6/4